

Allied Systems: Data governance challenges and the opioid crisis

Kevin McDermott 

Abstract

This teaching case based on real people and events describe a critical decision point during the nascent stage of entrepreneurial venture called Allied Systems Integration Solutions in May 2019. This data governance case explores the challenges with centralizing client data in an effort to help health practitioners cope with the opioid use crisis. Readers are asked to put themselves into the shoes of the protagonists who must make difficult operational decisions related to these data governance and behaviour management questions. This case is derived from participant observation of eight entrepreneurial mentoring sessions with the protagonist entrepreneurs. Detailed notes of participant observation sessions were maintained and qualitative data were analysed for the purpose of creating this business case. This case is intended for upper year undergraduate, or MBA courses. In particular, it should be used in curricula exploring the complexities of real-world data governance decisions, managing the trade-offs between operational efficiencies and data protection. This case would be most suited to courses in Management Information Systems, Database Management and Operations Management.

Keywords

Opioid crisis, data governance, data quality, data access, social enterprise, data privacy

Introduction

In early March 2019, Nithin Paul and Jacqueline Cole, two undergraduate students in their final year at Wilfrid Laurier University (WLU) in Brantford, Ontario walked to the front of a busy presentation room to deliver a presentation to panel of judges. Unbeknownst to them this presentation would send them on a path towards building a business and putting them on a path towards becoming social entrepreneurs.

Weeks earlier, Jackie and Nithin learned about the Criminnovation Challenge hosted by the Schelgel Centre for Entrepreneurship and Social Innovation (Wilfrid Laurier University, n.d.-b). Participants in the challenge were tasked with analysing a societal issue and designing a novel and financially sustainable solution (Wilfrid Laurier University, n.d.-a). Winners would receive US\$2500 to start their venture. Jackie and Nithin recognized that the opioid crisis was a significant social issue, so they got to work analysing this problem and preparing their solution. Something that they did not expect was that the key challenges that they would face related to data governance and

people's complex relationships with information about themselves.

The opioid crisis and social service agencies on the front line – the issue

In 2019, an opioid crisis was killing many individuals in Canada. From 2016 to 2018, the number of apparent accidental opioid related deaths more than doubled (Government of Canada, 2019). Brantford Ontario (a city with a population of 100,000 residents, about 100km south-west of Toronto) had the highest rate of hospitalizations for opioid

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overdoses of any city in Ontario (Gee, 2019) and the second highest in all of Canada (Moro, 2019).

Downtown Brantford also had a homelessness issue, which is a negative social determinant of health contributing to the opioid crisis. The Executive Director of a local emergency shelter, Tim Phip explained this relationship, ‘When I look at the clients who come into the shelter here, almost 100 percent of them indicate some kind of drug use . . . When people are homeless, and there’s not a lot of hope with things out there, [drug use] becomes really tempting’ (Moro, 2019). Across Canada, it was estimated that 35,000 people were homeless each night. With housing affordability declining, these numbers are increasing on a year-over-year basis (Brant News, 2016).

Jackie and Nithin had lived alongside this social issue for the last 4 years of their undergraduate programmes at WLU Brantford, they saw the social issues firsthand in the downtown core. For example, Freddy,¹ a homeless man who frequently utilized Brantford’s four emergency homeless shelters. He described his homelessness as stemming from an acquired brain injury while in high school. The behaviour changes associated with his injury made it ‘impossible for me to finish school and to hold down a job’. Freddy was frequently seen asking passers-by for money, or a ‘timmies card’ (a gift card for the popular Tim Horton’s restaurant).

Downtown Brantford has over a dozen social service agencies, including homeless shelters, addiction services, job training centres and mental health services, whose missions are to help individuals in need. While Freddy was able to clearly articulate his needs, others were less willing or able to articulate theirs. Some were ashamed of their homeless past or current behaviour and are unwilling to divulge their drug use for fear of social-judgement or criminal prosecution (Canadian Mental Health Association, 2018). Accordingly, because they are unwilling to divulge some of their drug-involvement they sometimes avoid reaching out to social service agencies to the detriment of their health and wellbeing. In the context of the opioid crisis, this is a life and death issue – if an individual misjudged a dosage or were given a drug that was stronger than expected, they could overdose. Opioid overdoses could be easily treated with nasal spray drugs such as Naloxone, avoiding fatalities (Chimbar and Moleta, 2018). But anti-overdose treatments like Naloxone can only be administered after the overdose, when the person was unconscious. For this treatment to be effective, drug users needed to be physically nearby to those with Naloxone kits, the most likely location for these kits are at social service agencies. But if individuals at risk of opioid overdoses avoid social service agencies for fear of disclosing their lifestyle, overdose deaths would likely continue unabated.

Jackie and Nithin learned from their Criminology professor, Dr Erin Dej that one of the hurdles that users of social service agencies face is the continual and frequent recounting of their medical and social histories. Jackie and

Nithin realized that things would work better for everyone if clients did not have to recount their histories each time they visited a social service agency. In fact, it might even save lives.

Changes to provincial funding for social services – the challenge

This problem is only going to get worse. The Ontario Provincial Government was in the process of reforming the funding model for many social service agencies. All users of social service agencies that received Provincial funding will be required to assess the ‘acuity’ of their clients. The new Health-Based Allocation Model (HBAM) specified that, clients would now need to be assigned a score that related to their perceived ‘complexity’. This acuity scale would include factors such as: the expected number of procedures and treatments needed, age, sex, socio-economic status and several other social determinants of health (Government of Canada, 2018). Measures of all factors must be collected and stored, and the acuity score was to be used for prioritizing who should receive care, and in what order. For example, if a particular client has complex mental health and addictions issues (such as bipolar disorder coupled with an addiction to opioids) and if they have overdosed twice in the last month, their high acuity score and should give them priority access to shelter beds at a homeless shelter.

At a policy level, the rationale for this new funding change was clear – social service agencies operated in the context of resource scarcity, so clients who are most in need should receive care quickly, while those whose issues are less severe should wait. At the operational level though, this was quite difficult to implement. St. Leonard’s Community Services was an addictions and mental health service provider in downtown Brantford. They have bail supervision and verification services and counselling related to addictions and mental health. St. Leonard’s had been a foundational social service in downtown Brantford for over 30 years. Traditionally, they triaged clients and used an internal client database to track their clients’ histories and services provided to them. With the HBAM funding changes, St. Leonard’s would be required to store personally identifiable client data, including: names, contact information, acuity score and a list of services provided. For St. Leonard’s, this change in work-flow was manageable, since they already store client data in an electronic database. However, this would be a more complex change for many homeless shelters which do not already collect client data.

By way of example, the city of Brantford coordinated a homeless shelter programme in partnership with four social service agencies: The Salvation Army, Rosewood House, Nova Vita and the Youth Resource Centre (Ruby, 2018). Each agency provided emergency shelter for homeless men, women and youth on an ad hoc basis. Beds were not reserved, and clients could utilize the available beds – or not – depending on their needs and preferences. Under the new funding

model, each client utilizing any of these four homeless shelter would need to be uniquely identified and their acuity (their relative need for service) and separately assessed by each of the four organizations. These homeless shelters did not track this information and would need to ramp up their organizational capabilities to adhere to the new funding model. None of them had dedicated Information Technology (IT) staff, significant IT expertise, nor the financial means to implement expensive IT systems. Because each agency separately received provincial funding, each would be required to assess and store acuity scores for their clients. Some clients would thus need to retell their medical and social histories to each homeless shelter on their first visits – effectively duplicating effort four times and leading to frustration among clients. More than thirty social service agencies operating in downtown Brantford, also faced this challenge.

Allied Systems integration solutions – the opportunity

Nithin and Jackie proposed a solution: a shared database that would be accessible (via the cloud) to each of the social service agencies operating in downtown Brantford. This would effectively allow agencies to use client data that was collected by their partner organizations to learn about clients who have never previously used their services. The idea was to reduce the barriers to care by ensuring that clients' acuity score would be assessed only once. Also, difficult discussions related to drug use and addictions could be left to specialist agencies (such as St. Leonard's) whose experienced counsellors could approach these conversations in a way such that it would be less likely to drive clients away from life-saving services.

Allied Systems centralized database and card system – the proposed solution

On 1 March 2019, Jackie and Nithin presented their centralized database idea and called their proposed business 'Allied Systems'. At the Criminovation challenge, they talked about how a centralized database would allow participating social service agencies to save time and resources by not duplicating acuity-testing efforts. They also described a novel card solution that would allow clients to quickly identify themselves at the food bank, or homeless shelters. Rather than having to identify themselves by name – showing government issued identification, Jackie and Nithin proposed that clients carry a near field communication (NFC) enable card. The identification card would be issued to clients upon their first visit to one of the participating social service agencies and clients would be expected to touch their card to a reader upon when visiting an agency or utilizing one of their services. Utilization tracking of services is not something that is done well among agencies in downtown Brantford, so

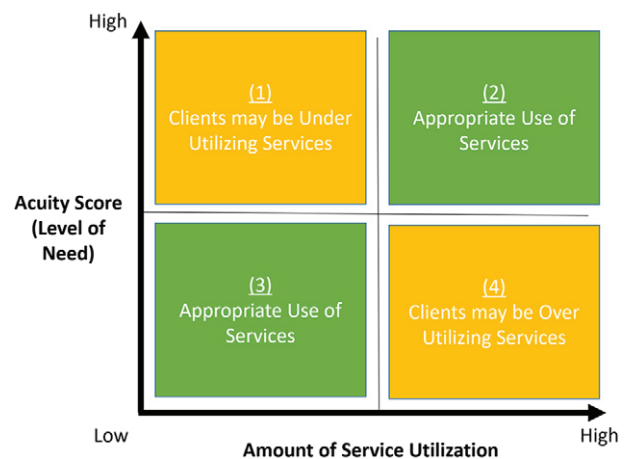


Exhibit 1. Matching clients with appropriate levels of service utilization.

Source: Created by the author.

the proposed system would provide some valuable metrics. Managers would be able to individually track and compare client acuity versus their utilization of services. This could help managers and clinicians determine which clients they should deprioritize when offering services and which clients should be proactively encouraged to utilize services (see Exhibit 1). Under the new HBAM funding model, agencies will only be required to assess client acuity, but Jackie and Nithin's proposed card solution provides social service agencies with more insight into their own system utilization. Managers would have an opportunity to reduce costs by deprioritizing services to over-utilizers in quadrant four, while improving health outcomes by actively targeting clients who are under-utilizers in quadrant one.

For this system to work, participating social service agencies would require some technology installed at each of their facilities. Specifically, they would need to have a computer or a mobile device with an NFC card reader located near the entrance of their facility or at their service locations. This NFC enabled device would need to be able to make a secure network connection to the centralized database. Finally, each client would need to be issued with an NFC identification card (see Exhibit 2).

Making this into a business – the decision

In addition to their presentation at the Criminovation Challenge, Jackie and Nithin also produced a one-page summary document describing their solution, see Exhibit 3. Their solution went over well with the judges at the Criminovation Challenge and among the competitive field of other proposals, their presentation was deemed the winning entry thereby earning the US\$2500 prize money. The competition was only an ideation exercise, so many operational details were not yet considered and would need to be

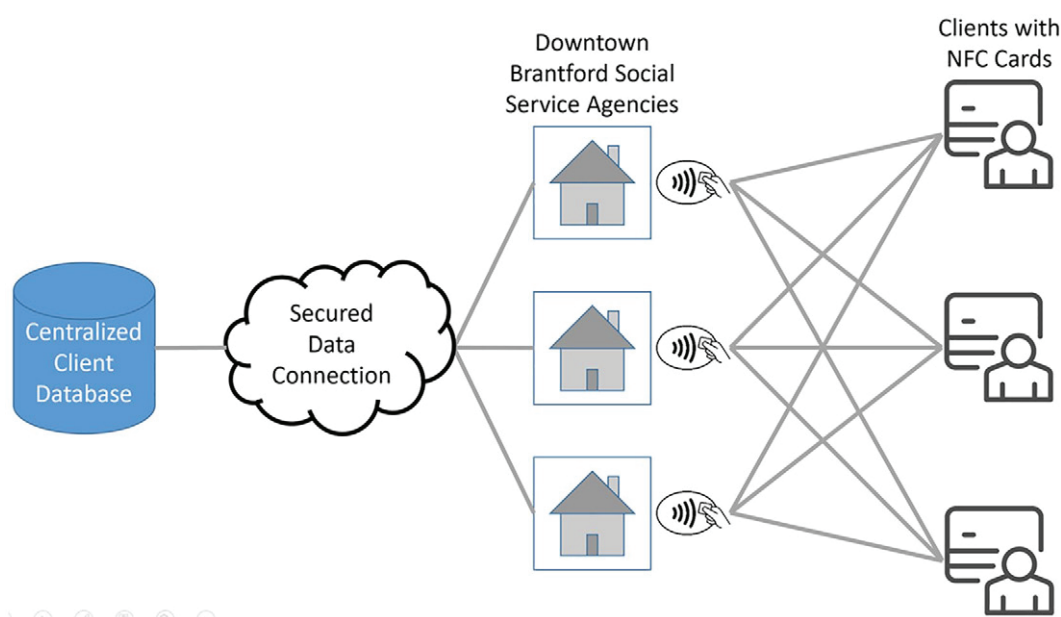


Exhibit 2. System diagram of proposed allied integration system.
Source: Created by the author.

thought through if their proposal were actually implemented. Accordingly, Jackie and Nithin came up with a business name – Allied Systems – and developed a logo and some branding material, see Exhibit 4.

With some momentum towards creating a real business, Allied Systems was inducted into the Laurier Brantford LaunchPad business incubator programme to provide them with an opportunity to put more thought into the idea. Once in the business incubator programme, they had access to resources to help them launch their new venture. These resources included access to further funding, business model advising and an advisory committee. Allied Systems was assigned a group of academic mentors who participated in monthly advisory committee meetings.

During their first meeting, one of their advisors Dr Kevin McDermott – Assistant Professor of Business Technology Management at WLU – while excited about the idea had some concerns. Dr McDermott taught Database Management to undergraduate business students and immediately identified several data governance related problems that could negatively impact the viability of the Allied Systems' proposal. Specifically, Dr McDermott saw three main concerns: (1) data ownership – who would own the data shared by the social service agencies; (2) data sharing and access – which organizations, employees and volunteers would have access to client data; and (3) data protection – how could Allied Systems ensure that client data is kept private and secure. It was clear that Allied Systems was proposing to store information that would personally identify clients, including full names and contact information alongside their social and medical acuity scores. The

possibility of deliberate or accidental release of this type of information could have harmful consequences on clients. Therefore, significant thought would need to be put into: what data should be collected; where it should be stored, and; how it should be protected.

Another member of the advisory committee, Dr Erin Dej, Assistant Professor of Criminology at WLU had extensive experience with homeless populations and conducted research on the type of clientele who utilize downtown Brantford social service agencies. She was an expert on homelessness, understanding the social determinants that lead to homelessness and its prevention. She too had some concerns with Allied Systems' proposal, they included: (1) were NFC cards really the best means of uniquely identifying clients – it seemed likely that cards would become lost or damaged through day-to-day street involvement and (2) many street or drug-involved individuals liked to remain anonymous and may be resistant to being tracked – potentially driving away high-needs/high-risk clients. If the system were to incorrectly assess the acuity of clients, or if drove away clients from potentially life-saving services, then the proposed system could have negative implications for the health of clients.

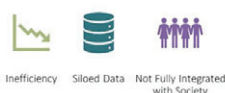
While well aware of the potential security and privacy issues, Jackie and Nithin considered some of the advisor's questions to be operational details that need not be considered at this point in their entrepreneurial journey. But upon further discussion with their advisors and through their own research they acknowledged that these issues could make or break their initiative. They would need to have answers to these questions prior to their next meeting – how should they address these issues?

CRIMINNOVATION

The Criminovation challenge enabled students to think critically of social issues in their community, and design solutions to address these issues. It was also aimed to connect students to faculty, mentors, and community representatives to assist them in the creation of their solutions.

We initially assumed that the citizens of Brantford were seeking employment but did not have the necessary means and resources to obtain work. After consulting with local businesses, community members and mentors, we discovered employment was not what the citizens of Brantford needed.

The problem was not that there was a lack of opportunities in the community, rather the social services within the city were not collaborating with one another, thus making it difficult for citizens to receive help and assistance to the fullest extent.



We designed a Systems Integration Solution. This is a proactive approach aimed to solve the separation and inefficiency that exist amongst the social service organizations within the Brantford area. This service is designed to connect all of the community organizations within the city to one another, so that they can collaborate and share user data to better serve the public. Through this system, organizations are not isolated from each other, which allows for the maximum utility of the limited resources of time, adequate finances, and the energy of the people involved. This approach streamlines care for citizens and prevents them from having to repeat their story each time they access a community service.

There are two main challenges to our solution; privacy and data sharing, and citizen and agency participation. Because this system operates on the basis of collecting data from citizens in the community, there is likely to be concerns over privacy. This is why our system relies on the voluntary submission of personal information. Another challenge is that both the agencies and the citizens of Brantford might not participate in the system. This system will take time to implement, so it is important that policymakers can understand the timeline and agenda of where the solution is heading. It is crucial that we are able to market our solution seamlessly so that all parties involved can see the benefits afforded to them by the use of our service.

It is conceivable that this approach could extend into other societal domains upon further research and development. It was a positive experience being able to work on a solution for social innovation. The opportunity to speak with faculty members and mentors was invaluable and we are grateful for the experience.

Exhibit 3. Criminovation challenge one page summary.
Source: Written and designed by Nithin Paul and Jacqueline Cole.



Exhibit 4. Allied Systems logo.
Source: Logo was designed and created by Nithin Paul and Jacqueline Cole.

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Note

1. This name is fictionalized to maintain anonymity for this real individual.

References

- Brant News (2016) Brantford, Brant counting homeless. *The Hamilton Spectator*, 29 March. Available at: <https://www.thespec.com/news-story/6410671-brantford-brant-counting-homeless/> (accessed 16 October 2019).
- Canadian Mental Health Association (2018) *Care not Corrections*. Available at: https://cmha.ca/wp-content/uploads/2018/04/CMHA-Opioid-Policy-Full-Report_Final_EN.pdf (accessed 16 October 2019).
- Chimbar L and Moleta Y (2018) Naloxone effectiveness: A systematic review. *Journal of Addictions Nursing* 29(3): 167–171.
- Gee M (2019) In a Brantford Tim Hortons, the toll of the opioid crisis is in full view. *The Globe and Mail*, 4 February. Available at: <https://www.theglobeandmail.com/canada/article-lessons-learned-from-a-downtown-brantford-tim-hortons/> (accessed 16 October 2019).
- Government of Canada (2018) *Canada Health Act Annual Report 2016-2017*. Available at: <https://www.canada.ca/en/health-canada/services/publications/health-system-services/canada-health-act-annual-report-2016-2017.html> (accessed 16 October 2019).
- Government of Canada (2019) *National Report: Apparent Opioid-related Deaths in Canada*. Available at: <https://health-infobase.canada.ca/datalab/national-surveillance-opioid-mortality.html#accidentalAORD> (", (accessed 18 October 2019).
- Moro T (2019) Brantford struggles to deal with opioid crisis and homelessness. *The Hamilton Spectator*, 28 March. Available at: <https://www.thespec.com/news-story/9244316-brantford-struggles-to-deal-with-opioid-crisis-and-homelessness/> (accessed 16 October 2019).
- Ruby M (2018) City launches homeless shelter system. *Brantford Expositor*, 16 December. Available at: <https://www.brantfordexpositor.ca/news/local-news/city-launches-homeless-shelter-system> (accessed 16 October 2019).
- Wilfrid Laurier University (n.d.-a) Criminovation competition. Available at: https://adc.wlu.ca/activedata/EventList.aspx?vview=EventDetails&eventidn=4495&information_id=13590 (accessed 18 October 2019).
- Wilfrid Laurier University (n.d.-b) Schlegel Centre. Available at: <https://students.wlu.ca/work-leadership-and-volunteering/entrepreneurship/index.html> (accessed 18 October 2019).

Author biography

Kevin McDermott is a professor of Business Analytics in the School of Business at Conestoga College in Kitchener, Ontario, Canada. Where he teaches, conducts research, and writes teaching cases. Prior to becoming an academic, he worked in the high-tech sector for over decade.